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MACH.2 and HAMR

We get the low down on HAMR and MACH.2 straight from the source

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Q Do you see the Indian consumers making intelligent buying decisions keeping in mind different performance characteristics of different HDD classes OR do they simply buy the drive within their budget?

Sameer: Indians are among the largest data consumers globally, and more conscious while investing in storage devices. The country is registering double digit growth in PC sales while the gaming industry is set to grow to \$1.1 billion by 2020. These trends paint an optimistic picture for the storage industry and are indicative of a shift in consumer buying behaviour. They understand the unique proposition behind each of the different HDD classes.

Q When are the Mach 2 and HAMR drives shipping for the consumer market?

Sameer: MACH.2 and HAMR drives are the future of capacity, one that is already here. We are preparing to ship 18 TB drives in the first half of 2020. We are also driving areal density leadership with our HAMR technology, which enables Seagate to achieve at least 20% areal density CAGR over the next decade. We remain on track to ship 20 TB HAMR drives in late 2020. As drive densities increase, multi-actuator technology is required to maintain fast access to data and scale drive capacity without compromising performance. We are working with multiple customers to qualify these drives, including Microsoft which is anticipating success with Seagate's Exos 2X14 enterprise hard drives with MACH.2 technology. Microsoft has completed its initial round of functional deployment testing of the Exos 2X14, as well as the tests

on the performance increase MACH.2 provides, which is the key advantage IT architects are looking for. We expect to see demand for dual actuator technology to increase as customers transition to drive capacities above 20 TB.

Q The latest Gartner study says that PC shipments have grown in Q3 2019 vs Q3 2018 (primarily laptop PC sales). What is Seagate's on this statistic?

Sameer: There is a surge globally in data creation and consumption. In India too, there is a strong growth not only on the consumer side but also enterprise with significant demand from data centres. While storage solutions catering to consumer needs – from everyday users and creative professionals to gamers – comprise a larger chunk of our business in India, we are witnessing an increase in demand from and consequently, a shift towards enterprise. Businesses are beginning to capitalise the potential of data, and hence, are investing in building robust data storage and management infrastructure.

Q What's the big deal about IT 4.0? What are Seagate's goals regarding the same?

Sameer: As more connected devices come onboard, we are staring at a data deluge. According to the Data Age 2025 study by Seagate, all data generated by 2025 will be an astonishing 175 zettabytes. More of this data will need to be analysed in real time. This exponential growth in data and the need for real-time analysis will contribute to the rise of the edge, which in turn, is the biggest driver behind IT 4.0. Edge computing will enable processing

of data at or near where it is collected, rather than on a cloud server, so analysis and answers can be provided quickly to make decisions and solve issues immediately. According to Gartner, while 91% of today's data is created and processed inside centralised data centres, by 2022 about 75% of data will need analysis and action at the edge. There are four reasons behind this shift: cloud-related latency; prohibitive cost of moving large data sets to the cloud; security threats to data; and network congestion.

What does this mean for businesses? For starters, they need to relook at their data strategy, building on their central cloud computing architecture to accommodate vast volumes of data for archiving and deep analysis & learning, while also developing the ability to process and securely store more data at the edge. It is clear that IT architecture of the past will not solve the challenges of tomorrow, which is why Seagate is helping build an infrastructure that supports the IT 4.0 era, and the demands of managing the burgeoning datasphere. By offering a unified data experience to businesses laden with 1s and 0s, Seagate is positioning itself to support this outcome. We'll do it through an array of solutions that work together and act as the instrumental layers between software and hardware so that data can be transported from end-points to the edge to the core seamlessly and securely. It's not only a framework that simplifies IT 4.0—it's the answer to consistent data protection, orchestration, and management for the industries, and people, needing it most. **Q**